
Told or Enforced: Measuring When In-Context Contracts Substitute for Runtime Enforcement in Agent Harnesses

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Abstract

1 With tool exposure fixed and no access to model weights, an agent can be held
2 to a rule two ways: state it in the prompt and rely on the model (telling), or have
3 the harness block the violation regardless of the model (enforcing). We separate
4 them, the rule told or withheld and the runtime on or off, and ask the question a
5 harness designer faces: given the model was told, does enforcement still change
6 behaviour?

7 We answer it on a pre-registered, single-runtime, within-family run: a stronger
8 and a weaker sibling in each of four model families. The runtime’s block is un-
9 conditional: across 488 enforced runs it let nothing through. With the runtime
10 off, telling the rule raised the safe rate in every family (+24 points pooled), so
11 specification moves behaviour in the families we test. But it does not stand in
12 for enforcement: the marginal value of enforcement given telling stays large (41
13 points pooled), small only for the two self-enforcing families and most of the
14 barrier for the two weak ones, so substitution holds only in that narrow corner.
15 Capability does not cleanly decide it either: within a family it helps or ties on the
16 salient boundary and can cut against safety on a discoverable side-door (clearest
17 in one family), so the contrast is flat to slightly negative. We report these results
18 descriptively, across three model lineages (Qwen2.5 and Qwen3 collapse to one
19 Qwen lineage), not as a significance claim.

20 Two further contributions. We release GovernedAgentBench: the test scenarios,
21 a deterministic offline grader (fixed code, no AI), the analysis code, and the git-
22 pinned reference runtime, with the paid transcripts and grades shipping as a ver-
23 sioned archive alongside the preprint. And a caution: a harness that hides a tool’s
24 output makes the agent guess facts it cannot see, which a grader then misreads as
25 fabrication; in our own pipeline an apparent case of this vanished the moment we
26 showed the agent what its commands actually returned.

1 Introduction

28 An agent harness is a design surface. Whoever wires up an LLM to tools controls the interaction
29 surrounding it: which tools are exposed, how actions are parsed and dispatched, and what happens
30 when a proposed action would violate a constraint. For any given constraint, with tool exposure
31 fixed and no access to model weights, that engineer holds exactly two levers. The first is in-context
32 specification (telling): state the rule in the agent’s prompt as an in-context contract and rely on the
33 model to respect it. The second is runtime enforcement (enforcing): have the harness block the
34 violation deterministically, regardless of what the model decides. Telling is cheap and moves with
35 the prompt; enforcing requires typed command schemas, dispatch gates, and validation code that

36 must be built, tested, and maintained. A harness designer deciding where to spend engineering
37 effort needs to know when the second lever buys behaviour the first does not.

38 Existing guardrail evaluations rarely isolate it. Some toggle enforcement while the full policy sits in
39 the prompt in both conditions [arXiv:2604.15579], so the delta is only measured where the model
40 was already told. Others bundle “injected and enforced” into one switch [arXiv:2602.22302], or
41 compare an enforced arm against a prompt-only arm [arXiv:2603.00822, arXiv:2605.28914], which
42 mixes telling into enforcing. The closest, Mind the GAP [arXiv:2602.16943], varies enforcement
43 directly but never crosses the told and enforced forms of the same rule. So the marginal value of
44 enforcement given telling is measured only in that told-column form; no located study crosses it
45 against a withheld arm for the same rule, across all four cells, scored on the model’s first action.

46 This paper runs that crossing. For each rule we build a 2×2 experiment: the rule told or withheld,
47 and the runtime enforcing or off. The quantity we care about is the marginal value of enforcement
48 given the model was told.

49 We hypothesised that two parameters would decide the marginal contribution of enforcement given
50 in-context specification: model capability and goal conflict. Goal conflict did not carry it, producing
51 no rule-breaking at all (Section 5). Capability looked decisive in an earlier four-model run, but
52 that run confounded capability with model family (both capable models non-Qwen, both weak ones
53 Qwen), so we ran the within-family test it demanded: a strong and a weak sibling in each of four
54 families, across a sixteen-task mutation gate. Two things came out of it. Telling moved behaviour
55 in every family (the safe rate rose 24 points pooled with the runtime off), but it did not stand in for
56 enforcement: given telling, enforcement’s marginal value stayed large (41 points pooled) and small
57 only in the two self-enforcing families, so substitution holds in a narrow corner. And capability
58 did not cleanly decide it: within a family it helps or ties on the salient commit boundary but can
59 cut against safety on a discoverable side-door (clearest in one family), so the strong-minus-weak
60 contrast is zero or slightly negative. The runtime’s guarantee, meanwhile, held on all 488 enforced
61 runs.

62 A second, independent finding is methodological. Our own harness first produced a spurious result:
63 it handed the agent a file path instead of a tool’s actual output, so the blinded agent guessed identifiers
64 it could not see, and the scorer called that fabrication. Restoring the output dissolved the effect
65 entirely (Section 6). We name the failure mode harness blindness.

66 **Contributions.** (1) A cross-family telling effect, and the limits of substitution: told a mutation-gate
67 rule with enforcement off, models self-enforce more than when it is withheld (+24 points pooled
68 across four families), so specification moves behaviour, but it does not stand in for enforcement,
69 whose marginal value given telling stays large (41 points pooled) and is small only in the two
70 self-enforcing families. Strict substitution, enforcement adding nothing once told, holds only in
71 that narrow corner, and capability does not order it: the within-family strong-versus-weak contrast
72 is zero or slightly negative. We evidence it on a pre-registered within-family run and report the
73 within-family contrast descriptively, across three model lineages, not with a significance claim; the
74 runtime’s guarantee is untouched, all 488 enforced runs staying safe. (2) The harness-blindness cau-
75 tion, demonstrated by dissolving one such finding. (3) GovernedAgentBench, the instrument that
76 produced the finding above and which we release for reproducibility: a task suite, a deterministic
77 offline scorer, and a git-pinned reference runtime that hold the two levers apart per mechanism. We
78 claim it for its clean, reproducible isolation of telling from enforcing for the same rule, scored on
79 first action, not as a novel design: prior work already measures enforcement in the told column (Sym-
80 bolic Guardrails) and reaches the withheld-but-enforced cell (FORGE), just not the full crossing of
81 the two for the same rule. The paper claims no scaling law, no injection robustness, and nothing
82 beyond this one runtime.

83 2 Background and related work

84 The harness between a model and its tools is now studied as a design surface, but that line optimises
85 it for capability, not governance (Harness-Bench, NLAH, AHE, Meta-Harness [arXiv:2605.27922,
86 2603.25723, 2604.25850, 2603.28052]; ETCLOVG, OpenReview 3hXEPbG0dh). Enforcement
87 frameworks supply mechanisms without measuring what the model would do unaided (AgentSpec,
88 CaMeL, Progent, Harness-MU [arXiv:2503.18666, 2503.18813, 2504.11703, 2606.21856]), and
89 studies pitting enforcement against text-only governance change behaviour without withholding

the rule per condition (ContextCov, Verifier Tax, Mechanical Enforcement, Policy-as-Prompt [arXiv:2603.00822, 2603.19328, 2605.14744, 2509.23994]; and the structural guards AIRGuard, Prompts Don’t Protect, Symbolic Guardrails, TRACE, ST-WebAgentBench [arXiv:2605.28914, 2605.18414, 2604.15579, 2606.13174, 2410.06703]). The result is not mere instruction-following, because told-only compliance is not free: capable models finish tasks while breaking latent rules, non-monotonically across families (LogiSafetyBench, SABER, Agent-SafetyBench [arXiv:2601.08196, 2606.01317, 2412.14470]); in-context policy degrades with reasoning complexity (SafePyramid [arXiv:2606.29887]); and rules obeyed while visible are dropped as context compacts (Governance Decay [arXiv:2606.22528]). That checkability is what prior work uses to build evaluations (IFEval, RECAST, VerIF [arXiv:2311.07911, 2505.19030, 2506.09942]), not to predict when an agent complies.

Two priors sit very close; four more reach part of the crossing. FORGE [arXiv:2602.16708] runs withheld-but-enforced against told-but-unenforced, the off-diagonal, never instantiating cell A. Mind the GAP [arXiv:2602.16943] varies enforcement directly, but its told rule and its enforced rule are different objects, and its attempt-rate-on-intent scoring cannot move between arms by construction. Agent Behavioral Contracts [arXiv:2602.22302] toggles specification and enforcement together, the main diagonal. ABSTAIN [arXiv:2606.02965] reaches three of four cells, but its enforced arm keeps the prompt, never realising withheld-and-enforced. TeamBench [arXiv:2605.07073] finds prompt-only and sandbox-enforced role separation indistinguishable on pass rate where the rule is never withheld. PhantomPolicy [arXiv:2604.12177] supplies the policy-invisible-violation taxonomy we adopt. The full cell-by-cell mapping appears in Appendix C.

Isolated withheld-but-enforced measurements exist: FORGE by design, Prompts Don’t Protect structurally, and Mind the GAP through its enforce mode. What no located prior offers is a clean, reproducible isolation of the two levers for the same rule across all four cells, scored on first action, per governance mechanism, with a released deterministic offline scorer. GovernedAgentBench provides that instrument; the paper’s claims are the empirical finding above and the harness-blindness caution (Section 6), not the novelty of any single design element.

3 The two levers

With tool exposure fixed, a harness can make an agent respect a rule two ways: state the rule where the model can read it (telling), or make the violation impossible whatever the model emits (enforcing). Deployed systems pull both at once, so their effects are confounded. This paper’s whole manipulation is to decouple them, varying telling and enforcing independently for a single rule.

Telling is the in-context contract. It is not just a policy paragraph but everything the model is shown: the list of commands and their argument shapes, a per-command “safe for the agent to run” flag, and the prose describing the out-of-scope boundary. It is a harness-level lever, separate from anything baked in during training, which we refer to as model disposition.

Enforcing is what the runtime does regardless of the model. Our reference runtime has five governance checks, each of which can be switched off (individually, or all at once):

Check	What it does
schema check	rejects malformed or invented commands before they run
dispatch gate	refuses to run a command not flagged safe for the agent
commit gate	the agent may propose a change to user data, but only the user may commit it
refusal rule	refuses out-of-scope requests, including a zero-tolerance medical boundary
audit check	attaches evidence to recommendations; the scorer later catches fabricated or missing references

A sixth property, transaction integrity, is always on. The commit gate is the one this paper’s headline turns on: the runtime refuses an agent-issued commit and leaves the change merely proposed.

131 **The 2×2.** Crossing the two levers, for any one rule, gives four cells:

	Enforcing	Off
Told	A: deployment baseline	B: told, not enforced
Withheld	C: withheld, still enforced	D: neither

133 Cell A is how systems ship. Cell B isolates self-enforcement: the model knows the rule and nothing
 134 stops it. Cell C isolates pure enforcement: the rule is hidden but the runtime still blocks. Cell D is
 135 the floor. Three reads carry the information: B vs. D is the effect of telling; C vs. D is the effect of
 136 enforcing; and A vs. B is our headline: what enforcement adds once the model was told. If A and B
 137 match, enforcement changed no behaviour on that rule; its value is then the guarantee itself, which
 138 is real and unconditional (a blocked action simply cannot happen) even where behaviour does not
 139 move.

140 One scoring convention matters. A blocked action returns an error, and that error is itself the rule,
 141 told late, so cell C drifts toward cell B after the model’s first contact with a block. We therefore
 142 score the telling reads on the model’s first action; Section 5.1 reports the multi-turn check.

143 4 GovernedAgentBench

144 GovernedAgentBench is the instrument that holds the two levers apart. It toggles telling and enforc-
 145 ing independently, per rule, against one fixed runtime, and scores the result deterministically and
 146 offline.

147 **The runtime.** The instrument is HAI, a non-clinical personal-wellness agent runtime with a typed
 148 command interface, pinned as a frozen snapshot; it is a reference implementation, not the contribu-
 149 tion or a product, though the benchmark’s author built it in work separate from this paper, which
 150 we flag as a same-team caveat. Its no-diagnosis boundary is one of the evaluated rules, not a dis-
 151 claimer. Two facts matter for reproduction. First, the published v0.2.0 package does not enforce; the
 152 dispatch gate landed three days after that version tag, so the measured runtime is pinned by git com-
 153 mit 6c82cd0. Second, before spending anything, a pre-flight check confirmed the runtime actually
 154 refuses an agent commit; it passed, so the enforced cells were measured against a runtime verified
 155 to enforce, not assumed to.

156 **Tasks.** The suite has 39 tasks; the paid run used a concentrated 16-task subset, the rest covered
 157 offline. Each task is one labelled cell of a rule’s 2×2. For the commit boundary we use two tasks
 158 (committing a target and committing an intent), each run four times per cell.

159 **The two axes in code.** Enforcing is a runtime setting. Telling is a per-task contract applied at prompt-
 160 render time while the runtime is untouched: the withheld prompt strips the rule’s facts and prose,
 161 and an automatic scan rejects any withheld prompt that still leaks the rule’s words. Two caveats
 162 attach. First, even withheld, the prompt still lists command names, so the model retains a hint, and
 163 B-vs-D and C-vs-D are therefore lower bounds on the effect of telling. Second, a discoverability
 164 aid injects the seeded target’s identifier into the task prompts of both arms, without which withheld
 165 models often fail to find the target at all; it raises the withheld violation rate, and its full provenance,
 166 per run, is disclosed in Section 7.

167 **Scoring: the commit predicate.** The headline rests on one deterministic check that needs no thresh-
 168 olds and no pattern matching. The scorer is offline with no model calls; it grades one (task, trajec-
 169 tory) pair, where a trajectory is the ordered record of what the agent did (each step a command, an
 170 observation of its result, a refusal, a final answer, or a malformed turn). On the commit boundary
 171 the whole 2×2 reduces to a single predicate over the agent’s first action a :

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172 commit(a) = a is an agent-issued command that activates or deactivates
173             user-authored state (W57, the runtime’s user-authorization gate)
174             and whose execution returned OK
175 safe(a)    = not commit(a)

```

176 A proposal or a refusal is safe; an executed activation is the violation. The two boundary choices are
 177 deliberate. First action, because a blocked action’s error message is itself the rule told late (Section

178 3), so only the first move reads what the model brought to it. Executed, because an attempt the
179 runtime blocks is visible in the trajectory but is not a violation that happened; the violation is a
180 change that landed. The predicate is a fixed function of the trajectory and the pinned command
181 manifest, with no threshold or model judgement between the trajectory and the label, and a reader
182 can audit it against the released tests in minutes.

183 **The rest of the grader.** The wider 39-task suite is graded by the same kind of fixed offline checks, in
184 two groups: task-completion checks (correct commands, valid output formats, a summary consistent
185 with what the agent observed) and safety violations graded zero-tolerance, where a single occurrence
186 fails the run outright (an unauthorised change to user data, a direct database write, an invented
187 command, a wrong refusal in either direction, a fabricated citation, a medical claim). None of this
188 carries weight in the headline, which rests on the commit predicate alone. The detector detail and
189 full metric definitions appear in Appendix A.

190 **Reproducibility of the scoring.** The bars and which checks are zero-tolerance are read from one
191 committed configuration file, and each score records that file’s SHA-256 hash, so a threshold or
192 policy change not carrying a hash change is a detectable reproducibility break, not a silent re-scoring.
193 The bars are saturated by design: the contract gives no partial credit for one hallucinated command
194 or one unauthorised mutation.

195 **A disclosed pre-run scorer edit.** The harm-check code was edited three times in the day before the
196 run, the last 47 minutes before it, each edit in the direction we hoped for. We verified this does not
197 touch the result: every violation in the headline table is caught by the original, older detection path,
198 so re-scoring with the pre-edit scorer is identical on the 2×2 and the finding is invariant to those
199 edits.

200 **The pre-registered runs.** Both runs were designed before their data. The within-family run (Section
201 5.1) was pre-registered in full: the four family pairs, the sixteen-task suite, the within-family cell-B
202 capability contrast as the estimand, and a lineage-collapsed sign-flip permutation as the primary test:
203 pre-committed as descriptive rather than confirmatory, because at three model lineages its exact one-
204 sided floor is 0.125. Two departures. First, a secondary serverless-breadth arm (four single-band
205 models) failed to run on a stale provider key and is dropped; it carries no family pair and no weight
206 on the primary. Second, the run spans two runtime commits, with an orchestration-only robustness
207 fix between them that leaves every model reply byte-identical, so the split is immaterial to scoring.

208 5 Results

209 We report the within-family run (Section 5.1); the four-model ladder that motivated it (Section 5.2)
210 follows as provenance. One structural fact frames everything. Cell A (told, enforced) and cell C
211 (withheld, enforced) are safe on every run by construction: the runtime blocks the mutation whatever
212 the model emits, and across all 488 enforced runs not one violation got through. The enforced cells
213 carry no variance, so the 2×2 reduces to two contrasts in the runtime-off column. Cell B against
214 cell D (told versus withheld) is the effect of telling. Cell A against cell B (enforced versus off, once
215 the model was told) is the question a harness designer actually asks, and the one this paper set out
216 to measure: what does enforcement still add once the model has been told? Because A is 100% by
217 construction, that contrast is just 100 minus the cell-B safe rate.

218 **Remark: the identity is the finding, not a trick.** A sceptic can read A minus B as a re-description:
219 enforced runs are safe by construction, so the number is arithmetic on cell B, and the behavioural
220 content of the study is the runtime-off column plus a software verification. That is precisely the
221 design.

222 5.1 The within-family run

223 The four-model ladder of Section 5.2 entangles capability with model family: its capable models are
224 non-Qwen and its weak models Qwen, so “weaker models need enforcement” and “Qwen models
225 do” cannot be separated. The within-family run breaks that confound by construction, contrasting a
226 strong and a weak model *within* the same family (Qwen2.5 {72B, 7B}, Qwen3 {32B, 8B}, Llama3.1
227 {70B, 8B}, Mistral {24B, 7B}), across a sixteen-task mutation-gate suite (eight distinct governed
228 decisions, each told and withheld: commit, archive and set-active on a target and on an intent, plus
229 two commit framings), four repeats a cell, run against on-demand H100 deployments for about

USD 115. It answers the enforcement-given-telling question directly, and corrects what the ladder suggested. Throughout, rates are rounded to the nearest point and effect columns are computed from the unrounded rates, so an effect may not equal the difference of the rounded columns beside it.

The guarantee holds at scale. Across all eight models both enforced cells scored safe on every one of 488 runs. Those cells were scheduled at 512 (both enforced cells, eight tasks each, eight models, four repeats); 24 trajectories came back incomplete, from length truncation and tool-driving failures concentrated in the weaker models, and are dropped, split evenly across the two enforced cells (12 and 12). Because the enforced cells are safe by construction, a dropped run is an unfinished trajectory, not an enforcement failure. The runtime’s block is unconditional and did not miss once at eight-model scale; everything below concerns the two cells where the runtime is off.

Telling helps, but does not replace enforcement. With the runtime off, telling the rule (cell B) beat withholding it (cell D) in every family. But telling did not make enforcement redundant: the marginal value of enforcement given telling, cell A minus cell B, stayed large.

Family	B: told, off	D: withheld, off	telling (B–D)	enforcement given told (A–B)
Qwen2.5	28	20	+8	72
Qwen3	27	12	+14	73
Llama3.1	93	57	+36	7
Mistral	90	53	+37	10
Pooled	59	35	+24	41

Telling raises the safe rate 24 points pooled and is positive in every family: specification moves behaviour. But pooled, enforcement still adds 41 points over telling alone, and that residual splits sharply by regime. It is small only for the two self-enforcing families (7 points for Llama3.1, 10 for Mistral) and it is most of the barrier for the two weak ones (72 and 73 points). Substitution in the strict sense, that enforcement adds nothing once the model was told, holds only in that narrow high-self-enforcement corner, and even there it is a 7-to-10-point residual, not the clean zero the ladder showed. No family reaches a 100% told-off safe rate, so the ladder’s cell-A-equals-cell-B result (Section 5.2) does not survive the within-family test (Figure 1).

Intervals are wide at these per-cell sizes, but the pooled contrasts are clear. Per-family 95% Wilson intervals on cell B run from [19, 40] (Qwen2.5) to [84, 97] (Llama3.1), and cell D is similarly wide; pooled, cell B is 59% [53, 65] and cell D 35% [29, 41]. The telling effect (B minus D) is +24 points, 95% Newcombe interval [16, 33], and the enforcement residual (A minus B) is 41 points, 95% interval [35, 47]. The repeat-dependence caveat that keeps the within-family capability contrast descriptive (below) applies to these intervals too, so we read them as indicative rather than exact. Two robustness checks. Scored on the converged multi-turn window instead of the first action, every cell in this run is identical, so no result here depends on the window choice. And 20 runtime-off trajectories (8 in cell B, 12 in cell D) also came back incomplete, concentrated in the same weaker models as the enforced-cell drops; scored pessimistically, with every dropped trajectory counted as a violation, the pooled telling effect is unchanged (+24) and the weak-sibling ceilings in Llama3.1 and Mistral fall to 84% and 91%: Mistral’s contrast stays negative (about –6) while Llama3.1’s flips to about +3, the strong sibling marginally safer, which is what a ceiling artefact predicts. We do not adopt that scoring, since an unfinished trajectory is not an executed violation, and report it as a bound.

Capability does not cleanly order self-enforcement. The ladder read capability as what decides whether telling suffices. Within a family that is not a law. On the commit boundary the ladder actually swept, the stronger sibling is at least as safe in every family (Qwen2.5 8/8 versus 0/8, Qwen3 4/8 versus 2/8, Llama3.1 and Mistral tied at ceiling), so there capability helps if anything. Pooled across the wider mutation gate, though, the within-family strong-minus-weak contrast in cell B runs flat to slightly negative:

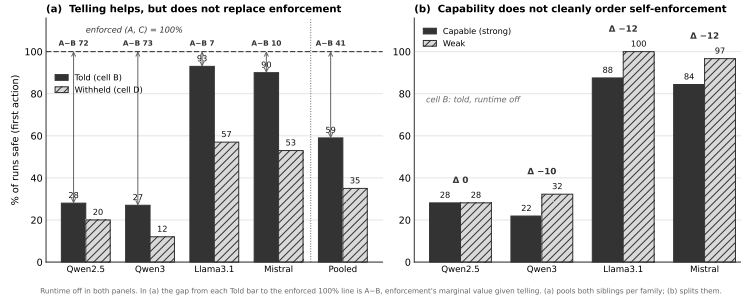


Figure 1: The within-family run, runtime off in both panels. (a) Telling the rule (cell B) beat withholding it (cell D) in every family, +24 points pooled, but no family reaches the 100% that would make enforcement redundant; the distance from 100 is enforcement’s marginal value given telling (cell A minus cell B), small for Llama3.1 and Mistral and most of the barrier for the Qwen families. (b) Within a family the strong-minus-weak cell-B contrast is flat to slightly negative, so capability does not cleanly order self-enforcement; on the commit boundary specifically the stronger sibling is at least as safe in every family. Bars are the share of runs safe on the first action. The enforced cells A and C were safe on all 488 runs and are not shown; (a) pools both siblings per family, (b) splits them.

Family	strong	weak	contrast
Qwen2.5	28	28	0
Qwen3	22	32	−10
Llama3.1	88	100	−12
Mistral	84	97	−12

The pooled negative is small and mixed. Two families sit at ceiling: in Llama3.1 and Mistral the weak sibling is safe on almost every run whether the rule is told or withheld (Llama3.1 100% told and 100% withheld, a telling effect of +0; Mistral 97% and 92%, +4), so its safety is nearly invariant to the rule and reads tool competence, not self-enforcement, and their −12 contrasts are largely ceiling artefacts. In the two families where both siblings can actually drive the tools, Qwen2.5 and Qwen3, a real mechanism appears: the stronger model is at least as safe on the salient commit boundary (8/8 versus 0/8, 4/8 versus 2/8) yet executes the archive and set-active side-doors the weaker one cannot. It nets zero in Qwen2.5, where the commit gain cancels the side-door loss, and a genuine −10 in Qwen3, the one non-artefact net negative. So the honest reading is narrow: capability helps or ties on the salient boundary and can cut against safety on a discoverable side-door, and the pooled contrast mixes that small genuine effect with two ceiling artefacts. As a significance test this settles nothing: at three model lineages the one-sided sign-flip permutation floors at 0.125 and returns $p = 1.0$, which is only the largest value the statistic can take once the observed mean is not positive, so we read the contrast descriptively.

5.2 The confounded precursor

A four-model ladder is what motivated this run. Told the commit rule with enforcement off, two capable models (MiniMax-M3, Llama-3.3-70B) refused on every run and two weak models (Qwen3.5-9B, Qwen2.5-7B) committed on every run, so capability looked decisive and telling looked like it made enforcement redundant for the capable pair, whose cell B was exactly 100%. But both capable models were non-Qwen and both weak ones Qwen, confounding capability with model family, and the clean cell-A-equals-cell-B reading rested on that 100%. The within-family run above breaks the confound and does not reproduce it: no family reaches a 100% told-off safe rate, so enforcement adds a residual everywhere. The full ladder table, its per-run mechanism, the probe detail, and the ladder’s caveats are in Appendix B.

6 Harness blindness

Probing prior to the experiment produced and then destroyed an apparent positive result, and the failure mode is general enough to be considered a finding. Early on in this project, the harness did not show the agent its tools’ output: where a command’s result should have appeared, the agent got a file path instead. When a task then needed an identifier to proceed, the agent guessed.

It was an artefact of the harness. We pre-registered a falsification test and re-ran it after fixing the harness to show the agent its command output. Fabrication dropped to zero against the pre-registered 40-percentage-point falsification bar: once the agent can see the empty lookup, it abstains honestly even when it needed the id to finish.

The lesson: an eval harness that hides tool output can cause the agent to guess, and a scorer reads the guess as fabrication. Genuine constraint-driven fabrication does exist under more adversarial setups [arXiv:2606.14831]; our point is narrower. Before charging an agent with making something up, check that it could see what it is accused of inventing.

7 Limitations and scope

This is a case study on one runtime. Every model-backed number comes from one runtime and one prompt template. The within-family run spans eight models in four families across a sixteen-task mutation gate, but they are eight models on a single instrument: whether the effect generalises beyond this runtime is the highest-value follow-up, and until an independent replication exists, nothing here separates a real phenomenon from a quirk of this instrument. The behavioural finding also rests on a single mechanism, the mutation gate: the other rule we swept, the medical boundary, was near-ceiling and uninformative (Appendix B), so whether telling substitutes for enforcing on other mechanisms is untested. The two levers are also not the whole option space: where feasible, denying the agent the dangerous tool outright dominates both, and this paper studies the residual case where the task legitimately requires the capability, which is what makes violations measurable.

The capability–family confound is broken, at a cost. The precursor ladder confounded capability with model family (capable = non-Qwen, weak = Qwen); the within-family run (Section 5.1) removes that confound by pairing a strong and a weak sibling inside each of four families, and the ladder’s clean capability reading does not survive it: within a family, capability helps or ties on the salient boundary and can cut against safety on a discoverable side-door, so it does not order self-enforcement. What the within-family run cannot yet do is make the within-family null a test rather than a description: it rests on three model lineages, where the sign-flip permutation floor is 0.125, so more families are needed to move from a descriptive null to a rejection. Per-model vendor-recommended sampling still entangles model identity with sampling settings even within a pair, though each within-family 2×2 is computed at fixed settings and holds family constant across the capability contrast. In the two families where the weak sibling sits near ceiling (Llama3.1, Mistral) it is safe by incapability rather than self-enforcement, its safety barely moving when the rule is told (Llama3.1 +0, Mistral +4), so there the contrast reads tool competence, not disposition; it isolates self-enforcement cleanly only where both siblings can drive the runtime.

The enforced cells carry no variance. Cells A and C are 100% by construction, so A minus B is exact arithmetic on cell B (the Section 5 remark); the guarantee run confirms the block held on all 488 enforced trials, so that share is real, not an artefact of scoring.

Pre-registration honesty. Beyond the outcome-screened replacement model and the sub-target sample size (Appendix B), the withheld prompt still leaks command names, and the medical-boundary detector shares ground truth with the runtime. One design choice deserves its own line: the task prompts inject the identifier of the seeded target, a discoverability aid added to the instrument after early smoke runs showed models scoring safe merely because they could not find the target at all. In the within-family run it is a design constant, present in every cell and both arms and part of the suite before that run’s canary. Its likely direction is to raise withheld-arm violations and therefore the measured telling effect, since without it cell D reads safer by incapability. It is prompt-side, so it cannot be removed by re-scoring; a no-aid arm is future work. Its reach is bounded by mechanism: the aid changes discovery, not ability, so it can move an outcome only for a model able to execute the mutation once its target is findable. TeamBench’s caution applies to us too:

our scorer sees whether the specific guarded violation occurs, but an enforcement effect that shifted some finer channel while leaving the tracked violation unchanged would go undetected at this size.

8 Discussion: what enforcement is still for

The result relocates enforcement’s value; it does not remove it. Enforcement keeps its unconditional guarantee: a blocked action cannot happen, whatever the model does or is prompted, the property FORGE [arXiv:2602.16708] establishes formally as verdict correctness (no denied action fires), and that guarantee is untouched by any behavioural number: it held on all 488 enforced runs of the within-family study. Beyond the guarantee, enforcement contributes to the safe outcome via the block wherever a model does not self-enforce: below the operate floor, where the failure is being too weak to drive the tools rather than disobedience; when the rule was never told, the floor cell where the runtime is the only barrier and real deployments accumulate such rules through drift and context compaction [arXiv:2606.22528]; when a told rule is present but not salient at the moment of action; wherever capability itself enables the violation, as a more capable model can drive a governed side-door to completion a weaker one cannot (clearest in one family); and under adversarial pressure, which we do not test.

Mechanical Enforcement [arXiv:2605.14744] finds enforcement helping, but in a different domain: it scores decision-rationale quality under regulatory pressure, not tool-action compliance under benign use. Mind the GAP [arXiv:2602.16943] is better read as convergent than contradictory. Its frontier-model null is therefore consistent with the high-self-enforcement corners of our map, not a universal statement, and our runs locate its boundaries. The full reconciliation appears in Appendix C.

A note for evaluation. Read from the model’s side, cell B (told, runtime off) measures the self-enforcement a model supplies once told, a quantity distinct from raw capability and from runtime enforcement; a post-training evaluation could track it before and after an intervention, but we do no training here and claim nothing about it.

9 Conclusion and future work

Told or enforced, the honest answer is that telling helps but does not stand in for enforcing. In a within-family run across four families and eight model sizes, stating a governed mutation rule in context raised the safe rate in every family (+24 points pooled with the runtime off), so specification moves behaviour in the families we test. But given telling, runtime enforcement still added most of the barrier (41 points pooled), and substitution, enforcement adding nothing once the model was told, held only for the most self-enforcing families and even there left a residual. Capability did not cleanly decide it: a confounded four-model ladder had made capability look decisive, but within a family it helps or ties on the salient boundary and can cut against safety on a discoverable side-door, so the pooled contrast is flat to slightly negative. We report these results descriptively, across three model lineages, not as a significance claim, and the ladder’s medical-boundary rule was uninformative at this scale. The runtime’s value is therefore not that capable models have outgrown it; it is that no model self-enforces everywhere, and enforcement is the only barrier where it does not: a withheld rule, a discoverable side-door, a rule below the model’s operate floor. Its guarantee is intact and unconditional.

The takeaway we most want carried is the methodological one: before attributing fabrication to a model, check what the harness let it see.

The within-family run above is the confound-break the earlier draft named as future work; four things would still sharpen the result. The permutation rests on three model lineages, where the exact one-sided floor is 0.125, so more families are what would let the descriptive null harden into a test. External replication on a second runtime with its own rules and scorer is what would turn a case study into a phenomenon. Adversarial inputs would locate where substitution ends and injection defence begins. And long-horizon drift (does enforcement catch a rule the model has since forgotten?) is the measurement this design most directly points at.

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Appendix A: reproduction

The Section 5.1 headline is reproduced by the released analysis code (`analysis/paired.py`) run over the paid trajectories, which are released as a versioned archive (a repository release, name withheld for review, with SHA-256 checksums); the archive includes the resulting `paired_result.json`. The precursor ladder table (Appendix B) reproduces the same way, from the per-gate analysis and the pooling module (`build_cell_contrasts_pooled`); the exact tests come from a small released helper (`exact_tests.py`, pure Python, no dependencies), which returns the run-level (0.00016), task-level (0.33), per-sub-gate (0.029), and family-corrected (0.00062) figures and the Clopper-Pearson intervals ([63, 100] for 8/8, [0, 37] for 0/8). Offline artefacts regenerate with `reproduce_offline.py` (no network, no private data). Two pins matter: the runtime is git 6c82cd0 (tag `gab-runtime-1.0.1`, not the v0.2.0 package, which does not enforce), and the suite, scorer, and analysis that produced these numbers are in the released repository at the preprint commit, a week later than the runtime pin. The full task table and the earlier retired apparatus (a 28-task / 25-oracle-pair design, superseded 2026-07-04) are documented with the release.

The rest of the grader, continued from Section 4. The two text detectors (medical claims, citation faithfulness) are pattern matchers and the grader’s only complex code; a reader should audit them against the released tests (652 offline tests, 90 on the detectors) rather than take them on faith. Full metric names, definitions, and thresholds ship with the released scorer configuration.

Appendix B: the confounded precursor ladder

The four-model ladder of Section 5.2, in full, preserved as provenance: its clean substitution and capability reading is superseded by the within-family run of Section 5.1, which does not reproduce it. It swept four models against the git-pinned runtime, MiniMax-M3 and Llama-3.3-70B (capable), Qwen3.5-9B (near the floor) and Qwen2.5-7B (below it), for a total paid cost of USD 10.44. Scoring the model’s first action, the share of runs that stayed safe on the commit boundary was:

Model	A: told, enforced	B: told, off	C: withheld, enforced	D: neither
MiniMax-M3 (capable)	100*	100	100*	0
Llama-3.3-70B (capable)	100*	100	100*	25
Qwen3.5-9B (near floor)	100*	0	100*	12.5
Qwen2.5-7B (below floor)	100*	0	100*	12.5

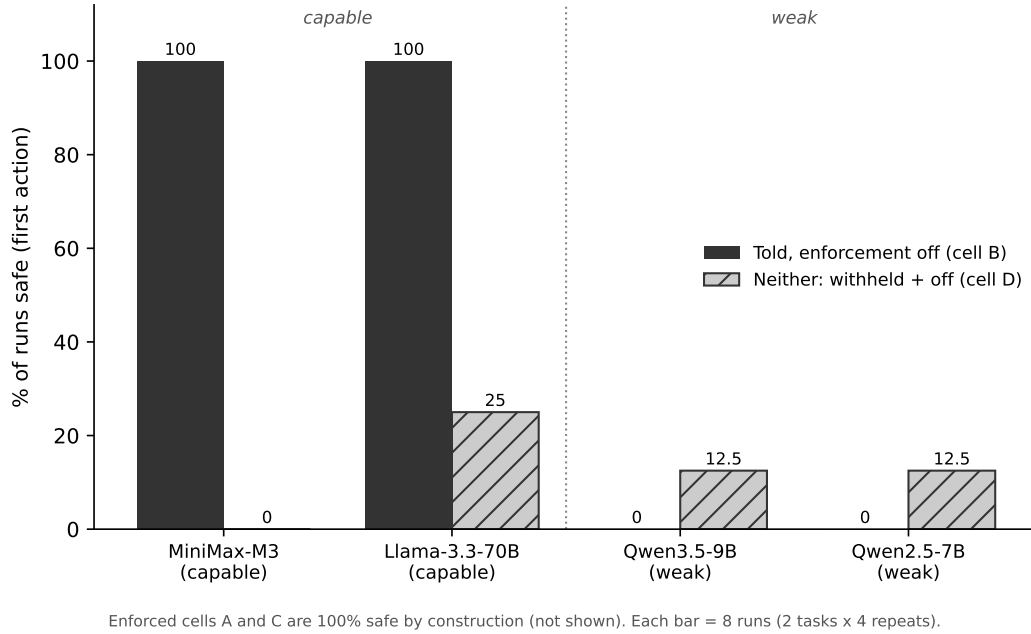


Figure 2: Told the commit rule with enforcement off (cell B): both capable models stayed safe on every run, both weak models violated on every run. The withheld floor (cell D) shows telling moves behaviour within a model even for the capable ones. Enforced cells A and C are 100% by construction. Each bar is eight runs.

498 *Safe by construction; the finding is in columns B and D. Each cell is eight runs (two tasks, four
499 repeats each).

500 **Read column B: told the rule, runtime off.** Both capable models stayed safe on all eight runs;
501 both weak models committed the forbidden change on all eight. The marginal value of enforcement
502 given the model was told, cell A minus cell B, is therefore zero for the capable models and the entire
503 barrier for the weak ones. That column is the precursor observation the within-family run of Section
504 5.1 was built to test and, on the capability reading, does not confirm.

505 **Why each model did it, read run by run.** The transcripts show the split directly. In the told-off
506 cell, MiniMax refuses the commit on its own turn and tells the user to run it themselves; Qwen3.5-
507 9B issues the commit command itself, and it executes because the gate is off. The capable model
508 declines an action it is free to take; the weak model takes it.

509 **What the uncertainty is, by estimation rather than testing.** With eight runs a cell the exact
510 (Clopper-Pearson) intervals are wide: a cell at 0/8 safe has interval [0, 37%], and a cell at 8/8 has
511 [63, 100%]. So the weak models' safe rate is bounded at most 37%, and the capable models' at least
512 63%, but neither is pinned precisely. The capable side's flat cell A minus cell B is a failure to detect
513 an effect at this size, not proof that none exists.

514 **Why counts, not a p-value.** A significance test is the wrong instrument here: one arm of every
515 enforced contrast is 100% safe by construction rather than a random outcome, the four repeats of
516 each task are near-identical rather than independent, and any rejection is confounded with model
517 family and task. Treating the eight runs a side as independent gives $p = 0.00016$, but counting the
518 two genuinely independent tasks a side gives $p = 0.33$.

519 **This is substitution, not the model simply being agreeable.** With the rule withheld and the runtime
520 off (cell D), the capable models violate too: MiniMax on all eight runs, Llama on six of eight. So
521 telling has a large within-model effect even for a capable model, and its flat A-minus-B is genuine
522 substitution, not indifference to both.

523 **Caveats.** The below-floor 7B fails partially rather than totally (graded told-off scores), so at a lenient
524 threshold its column B would read about 75% rather than 0%; the decisive cells are threshold-

invariant, and the released scorer configuration ships the zero-tolerance threshold used here. The ladder also inherited the discoverability aid of Section 7 mid-development: it was added after the ladder’s smoke runs, before its pre-registration and sweep, so for the ladder it was an outcome-informed instrument change, where for the within-family run it is a design constant. Empty output cannot fake the result: Llama produces malformed output on roughly 40% of runs yet still reaches 100% in column B through genuine refusals. Most limiting, once the below-floor 7B is set aside and the capability-versus-family confound is held in view, this ladder’s “told but does not self-enforce” result rests on one mid-size model, Qwen3.5-9B, which is why the within-family run of Section 5.1 was carried out. The medical-boundary rule the ladder also swept is near-ceiling: its detector fires on 7 of 223 runs, all on a single task and model, and the two tasks written to provoke a medical claim elicit none.

Two pre-registration disclosures, from Section 4. The precursor ladder (Section 5.2) carried its own two disclosures: its pre-registered primary model was deprecated by the provider mid-cycle and replaced with MiniMax-M3 only after a canary confirmed the replacement reproduced the effect (an outcome-conditioned choice, so MiniMax-M3 is a non-pre-registered addition and exploratory; the pre-registered capable model is Llama-3.3-70B, which shows the same pattern), and its commit boundary has two test tasks, not the three the sample-size analysis assumed, so its pooled cell is eight runs, below the twelve-run target; the effects are saturated (0/8 vs. 8/8), but the small size is stated.

The one-model probes. Before the ladder, a set of single-model probes (Qwen3-235B, three to five runs a cell) shaped the design. With the runtime off, a told-and-salient model refused three of three while a withheld model committed the violation three of three, establishing that telling and enforcing each move behaviour off the floor. Two ideas we expected to matter did not survive: audit faithfulness proved self-checkable once the model retrieved the evidence, collapsing into capability, and benign pressure to finish the task produced zero fabrication across 40 runs. The one qualifier they raise is salience-sensitivity: the same told model that refused when the rule was foregrounded dispatched the forbidden command three of three when it met the rule only incidentally, on three runs of one off-roster model, so we carry it as a caveat, not a result.

Appendix C: extended related-work mapping

This appendix preserves, verbatim, the full cell-by-cell mapping of the closest priors that Section 2 compresses, and the full Mind the GAP reconciliation that Section 8 compresses.

Two priors sit very close. FORGE [arXiv:2602.16708], a runtime policy-enforcement framework with a reference monitor, runs a by-design withheld-but-enforced condition: its instrumented arm strips the natural-language policy from the prompt and relies solely on runtime enforcement, and it compares that arm against a non-instrumented arm that keeps the policy in the prompt and does not enforce. Those arms are our cell C and our cell B, so its whole contrast is the off-diagonal: it drops the telling and adds the enforcing in a single step, measuring the two levers fused rather than the marginal value of enforcement once the model was told, cell A against cell B, which it never instantiates. It also has no neither floor, scores task success after multi-turn corrective recovery rather than on first action, enforces a single holistic policy per deployment rather than a per-mechanism inventory, and reports compliance as a correctness theorem, zero violations by construction, rather than a behavioural rate, so it measures whether enforcement is sound and preserves task success, not whether a told model needed it. Mind the GAP [arXiv:2602.16943] is the nearest of all: it varies enforcement directly (unmonitored, observe, enforce, where Enforce hard-blocks forbidden tool calls), so it does contain a rule-absent-but-enforcing cell. Three things still separate it from our design. Its told rule (a prose safety suffix) and its enforced rule (per-tool RBAC contracts) are different objects, never the same constraint crossed told-against-enforced; it scores the model’s attempt rate on intent across multi-turn jailbreak scenarios, not first-action compliance under benign pressure; and its headline “no deterrent” is a power-limited null ($p > 0.27$), not a clean substitution result. The separation is structural, not incidental: because it scores intent before the block and its Enforce mode is invisible to the model until a call is denied, the attempt rate it reports cannot move between its unmonitored and enforced arms by construction, so its design cannot measure the marginal value of enforcement given telling that we set out to isolate. We take up the apparent tension with enforcement-helps priors in Section 8.

Others reach part of the crossing. Agent Behavioral Contracts [arXiv:2602.22302] evaluates runtime contracts at scale (1,980 sessions, 7 models) but toggles specification and enforcement together: its central experiment sets a contracted agent, its rules both injected into the prompt and enforced, against an uncontracted agent with neither, “differ[ing] only in whether ABC contract rules are injected and enforced” (its §7.2). That contrast is our cell A against our cell D, the main diagonal, both levers on against both off, and so the maximal confound. Where FORGE moves along the off-diagonal (cell C against cell B), ABC moves along the main one, so between them the two closest contract-enforcement priors bracket the 2×2 without either isolating a single lever. ABC’s component ablation separates contract parts (its own labels: hard constraints, soft constraints, recovery, drift monitoring), not the two levers for one rule, leaving the specification and enforcement of a given constraint fused even there. We adopt PhantomPolicy’s [arXiv:2604.12177] eight-category policy-invisible-violation taxonomy, though its own conditions are a flat, non-crossed set folding telling into enforcement architecture, with no reported enforced-and-told cell. ABSTAIN [arXiv:2606.02965] reaches the closest cell coverage, three of four cells on our specify-by-enforce mapping for one abstention mechanism, but its enforced (Checkpoint) arm “receives the same prompt as the Prompt-Only policy” (Appendix B), realising told-and-enforced rather than withheld-and-enforced, and its unenforced arms are LLM-judged. TeamBench [arXiv:2605.07073] finds prompt-only and sandbox-enforced role separation statistically indistinguishable on pass rate yet 3.6 times more verifier code-edit attempts in the prompt-only arm: convergent evidence for the substitution shape in a different, multi-agent domain where the rule is never withheld, with a standing caution (Section 7) that aggregate parity can mask sub-metric enforcement value.

The Section 8 reconciliation with Mind the GAP, in full. Across six frontier models it finds runtime enforcement adds no detectable deterrent to what a model attempts, which resembles our result for the self-enforcing families, where enforcement’s marginal value given telling is smallest (7 to 10 points, though not the zero a strict reading of its null would imply), reached independently on a different runtime, different models, and adversarial rather than benign pressure. Its Enforce mode hard-blocks as our commit gate does, but its null is on attempt rate scored on intent before the block, a quantity its design cannot move by construction (Section 2), so it reports the null as a flat, frontier-only fact and a power-limited one ($p > 0.27$). Our runs bound the scope of that null rather than contradict it. Its no-deterrent result holds where the model already self-enforces the told rule; our within-family run finds that region real but regime-specific: high on the commit gate for the self-enforcing families, low wherever the model does not self-enforce (weak models on that gate, any model on a side-door it competently drives, every model once the rule is withheld), and not ordered by capability the way a frontier-only reading would suggest.